Mar Ephraem
College of Engineering and Technology

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A MINI PROJECT REPORT

on

"GAS LEAKAGE DETECTOR"

Submitted in partial fulfillment of the requirements
for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

A Gas Leakage Detector is a safety-oriented system designed to identify and alert users about the presence of hazardous gas leaks in residential, commercial, and industrial environments. The system primarily focuses on detecting gases such as Liquefied Petroleum Gas (LPG), methane, and carbon monoxide, which pose serious risks including fire hazards, explosions, and health issues.

This project utilizes a gas sensor module (such as MQ-series sensors) integrated with a microcontroller to continuously monitor gas concentration levels in the surrounding environment. When the gas level exceeds a predefined threshold, the system activates an alert mechanism, which may include a buzzer, LED indicators.

The proposed system is cost-effective, reliable, and easy to install, making it suitable for widespread use. It enhances safety by providing early detection and immediate warning, thereby helping to prevent accidents and protect human life and property. Additionally, the system can be expanded with IoT capabilities for real-time monitoring and remote alerts.

Overall, the Gas Leakage Detector serves as an efficient solution for minimizing risks associated with gas leaks and improving safety standards in various settings.

INTRODUCTION

In today's modern world, the use of gases such as Liquefied Petroleum Gas (LPG), methane, and natural gas has become an essential part of daily life in households, industries, and commercial establishments. These gases are widely used for cooking, heating, and various industrial processes due to their efficiency and convenience. However, despite their advantages, they pose significant risks when leaked unintentionally. Gas leakage can lead to dangerous situations such as fire accidents, explosions, and serious health hazards, which may result in loss of life and property.

One of the major challenges associated with gas usage is the lack of timely detection of leaks. In many cases, gas leaks go unnoticed due to the absence of proper monitoring systems or human negligence. Traditional methods of detecting gas leaks, such as relying on smell, are not always reliable and can be dangerous. Therefore, there is a critical need for an automated system that can continuously monitor gas levels and provide immediate alerts in case of leakage.

The Gas Leakage Detector system is designed to address this issue by providing an efficient and reliable solution for early detection of hazardous gases. The system uses gas sensors to detect the presence of harmful gases in the environment and a microcontroller to process the data. When the detected gas concentration exceeds a safe threshold, the system triggers an alert mechanism such as a buzzer or visual indicator, thereby warning the user instantly.

This project aims to develop a cost-effective, user-friendly, and highly responsive gas detection system that enhances safety in various environments. With advancements in technology, the system can also be integrated with Internet of Things (IoT) features, enabling remote monitoring and real-time alerts through mobile devices.

In conclusion, the Gas Leakage Detector plays a crucial role in ensuring safety and preventing potential hazards by offering a proactive approach to gas leak detection and management.

SYSTEM REQUIREMENTS

Hardware Requirements

The hardware components used in the Gas Leakage Detector system are responsible for sensing, processing, and alerting. The main components include:

- **Gas Sensor (MQ-2):**
  Detects the presence of gases such as LPG, methane, propane, and smoke in the environment.
- **Microcontroller (Arduino Uno):**
  Acts as the brain of the system. It processes the signals received from the gas sensor and controls the output devices.
- **Buzzer:**
  Provides an audible alert when gas leakage is detected.
- **LED Indicators:**
  Used for visual indication of system status (normal condition and gas leakage).
- **Power Supply:**
  Provides required electrical power to the system (battery or adapter).
- **Connecting Wires**
  Used to connect all components and build the circuit.
- **Relay Module (Optional):**
  Automatically switches off gas supply or electrical appliances during leakage.

Software Requirements

The software components are used to program and control the hardware system:

- **Arduino IDE:**
  Used to write, compile, and upload code to the microcontroller.
- **Arduino Programming Language:**
  Used for developing the logic to read sensor data and trigger alerts.
- **Operating System:**
  Windows / Linux / macOS for running the development environment.

Functional Requirements

The functional requirements define the key operations and behaviors of the Gas Leakage Detector system. These requirements describe what the system should do to ensure effective detection and safety.

1. Gas Detection

- The system must continuously monitor the surrounding environment for the presence of hazardous gases such as LPG, methane, and smoke.
- The gas sensor should detect gas concentration levels accurately.

2. Threshold Monitoring

- The system must compare detected gas levels with a predefined safety threshold.
- If the gas concentration exceeds the safe limit, the system should initiate an alert.

3. Alert Mechanism

- The system must activate a buzzer to provide an audible warning.
- LED indicators should glow to give a visual alert.
- The alert should continue until the gas level returns to a safe range.

4. Real-Time Monitoring

- The system should operate continuously in real-time without interruption.
- It must provide immediate response when gas leakage is detected.

5. Automatic Control (Optional)

- The system should automatically trigger a relay to shut off gas supply or electrical appliances during leakage.

6. System Initialization

- When powered on, the system must initialize all components and begin monitoring automatically.

7. Reliability and Accuracy

- The system should provide consistent and reliable performance under normal conditions.
- It must minimize false alarms and ensure accurate detection.

SYSTEM ARCHITECTURE

Overview:

The system architecture of the Gas Leakage Detector describes how different hardware and software components are organized and interact with each other to detect gas leakage and provide alerts. The system follows a simple embedded architecture consisting of input, processing, and output units.

1. Overview of Architecture

The Gas Leakage Detector system is divided into three main parts:

- **Input Unit**
- **Processing Unit**
- **Output Unit**

These units work together to continuously monitor gas levels and respond immediately when a leak is detected.

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2. Input Unit

The input unit consists of the **gas sensor (MQ series)** which is responsible for detecting the presence of harmful gases in the environment.

- The sensor continuously senses gas concentration in the air.
- It converts the detected gas levels into analog electrical signals.
- These signals are sent to the microcontroller for further processing.

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3. Processing Unit

The processing unit is the **microcontroller (Arduino Uno)**, which acts as the brain of the system.

- It receives input signals from the gas sensor.

- The analog signals are converted into digital values using an internal ADC (Analog-to-Digital Converter).
- The microcontroller compares the sensor values with predefined threshold levels.
- Based on the comparison, it decides whether the environment is safe or hazardous.

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4. Output Unit

The output unit provides alerts and takes necessary actions when gas leakage is detected.

- **Buzzer:** Produces an alarm sound to alert users.
- **LED Indicators:** Provide visual indication of normal or dangerous conditions.

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5. Data Flow in the System

1. Gas sensor detects gas concentration.
2. Sensor sends signals to the microcontroller.
3. Microcontroller processes and compares data with threshold values.
4. If gas level is normal → system remains idle.
5. If gas level exceeds threshold → alert system is activated (buzzer, LED, etc.).

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6. Optional IoT Integration

In advanced systems, an IoT module (such as Wi-Fi enabled NodeMCU) can be integrated:

- Sends real-time data to cloud platforms.
- Provides alerts through mobile applications or SMS.
- Enables remote monitoring of gas levels.

Architecture Diagram

```
 +-------------------+
 |   MQ-2 Sensor     |
 |  (Gas Detection)  |
 +---------+---------+
           |
           v
 +-------------------+
 |   Arduino Uno     |
 | (Microcontroller) |
 +----+--------+-----+
      |        |
LED Output   Buzzer Output
(Digital)    (Digital)
      |        |
      v        v
 +--------+  +--------+
 |  LED   |  | Buzzer |
 +--------+  +--------+
```

Key Features of Architecture

The architecture of the Gas Leakage Detector system is designed to ensure efficiency, reliability, and scalability. The following are the main features of the proposed architecture:

1. Layered Structure

- The system is divided into four layers: Sensing, Processing, Communication, and Application.
- Each layer performs a specific function, making the system easy to understand and manage.

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2. Real-Time Monitoring

- The system continuously monitors gas levels in the environment.
- It provides instant detection and quick response to gas leakage.

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3. Modular Design

- Each component (sensor, microcontroller, alert system) works as an independent module.
- Modules can be easily replaced or upgraded without affecting the entire system.

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4. High Reliability

- The system ensures accurate gas detection with minimal errors.
- Continuous operation improves safety and reduces risks.

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5. Scalability

- Additional sensors or modules (like temperature, smoke, or IoT devices) can be added easily.
- Suitable for both small-scale and large-scale applications.

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6. Efficient Data Processing

- The processing layer quickly analyzes sensor data and compares it with threshold values.
- Ensures fast decision-making and immediate alert generation.

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7. Multiple Alert Mechanisms

- Supports both audible (buzzer) and visual (LED) alerts.

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8. Low Cost and Energy Efficient

- Uses simple and affordable components.
- Consumes less power, making it suitable for continuous operation.

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9. Easy Maintenance

- Simple architecture allows easy troubleshooting and maintenance.
- Faults can be identified and fixed quickly.

IMPLEMENTATION

The Gas Leakage Detector system is implemented using a combination of gas sensors, microcontroller, alert mechanisms, and communication modules. The system continuously monitors the presence of hazardous gases and triggers alerts when gas concentration exceeds a predefined threshold.

1. Hardware Implementation

The hardware setup consists of the following components:

- **Gas Sensor (MQ-2):**
  Detects gases such as LPG, methane, and smoke. It outputs analog signals based on gas concentration.
- **Microcontroller (Arduino):**
  Processes sensor data and controls the overall system operation.
- **Buzzer:**
  Provides an audible alarm when gas leakage is detected.

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2. Software Implementation

The system software is developed using embedded programming:

- **Programming Language:** C/C++ (Arduino IDE)
- **IDE Used:** Arduino IDE
- **Libraries Used:**
  - Sensor handling libraries

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3. Working Process

1. The gas sensor continuously monitors the environment.
2. Sensor outputs analog data proportional to gas concentration.
3. The microcontroller reads the sensor values.

4. If the value exceeds a predefined threshold:
    - Buzzer is activated
    - LED turns ON (warning signal)
    - Relay turns OFF connected devices (if used)
5. If gas levels are normal:
    - System remains in monitoring mode

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4. Circuit Connection

- Gas sensor output → Analog pin of microcontroller
- Buzzer → Digital output pin
- LED → Digital output pin

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5. Algorithm

1. Start the system
2. Initialize sensor and modules
3. Read gas sensor value
4. Compare with threshold
5. If value > threshold:
    - Activate alarm
6. Else:
    - Continue monitoring
7. Repeat continuously

6. Code Implementation (Sample)

```
// Gas Leakage Detector using MQ-2, LED, and Buzzer
```

```
int mq2Pin = A0;      // MQ-2 analog output
int ledPin = 8;       // LED pin
int buzzerPin = 9;    // Buzzer pin
int threshold = 300;  // Gas level threshold (adjust after testing)

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int gasValue = analogRead(mq2Pin);
  Serial.print("Gas Level: ");
  Serial.println(gasValue);

  if (gasValue > threshold) {
    digitalWrite(ledPin, HIGH);
    digitalWrite(buzzerPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
    digitalWrite(buzzerPin, LOW);
  }
```

```
  delay(500);
}
```

7. Testing and Calibration

- The sensor is calibrated in a clean air environment.
- Gas is introduced near the sensor to test detection.
- Threshold values are adjusted based on sensitivity requirements.

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8. Deployment

- Installed in kitchens, gas storage areas, or industries.
- Mounted near potential leakage sources.
- Requires continuous power supply.

System Execution

The execution of the Gas Leakage Detector system involves real-time monitoring, processing, and alert generation. The system operates continuously to ensure safety.

Step-by-Step Execution Flow

1. **System Initialization**
   - Power supply is turned ON.
   - Microcontroller initializes all components (sensor, buzzer, LED, communication module).
2. **Sensor Activation**
   - Gas sensor starts sensing surrounding air.
   - It stabilizes after a short warm-up period.
3. **Data Acquisition**

    - Sensor continuously sends analog signals based on gas concentration.

4. **Data Processing**
    - Microcontroller reads sensor values.
    - Converts analog signals into digital values.

5. **Condition Checking**
    - Compares sensor value with predefined threshold.

6. **Alert Generation (If Gas Detected)**
    - Buzzer is activated.
    - LED glows (warning indication).
    - Relay (if connected) cuts off gas supply or power.

7. **Normal Condition**
    - If gas level is safe, system continues monitoring.

8. **Continuous Loop Execution**
    - Steps repeat continuously for real-time detection**.**

Features of Implementation

Key Features

- **Real-Time Gas Detection**
  Continuously monitors gas levels without interruption.
- **High Sensitivity**
  Detects even small amounts of gas leakage.
- **Automatic Alert System**
  Activates buzzer and LED instantly when leakage is detected.
- **Remote Notification**
  Sends SMS or app alerts using GSM/Wi-Fi module.
- **Low cost Implementation**
  Uses affordable and easily available components.

- **Easy Installation**
  Simple circuit design suitable for homes and industries.
- **Energy Efficient**
  Consumes low power during operation.
- **Scalable Design**
  Can be extended with IoT features and mobile applications.
- **Safety Automation**
  Relay module can automatically shut down appliances.
- **User-Friendly Operation**
  Requires minimal user interaction.

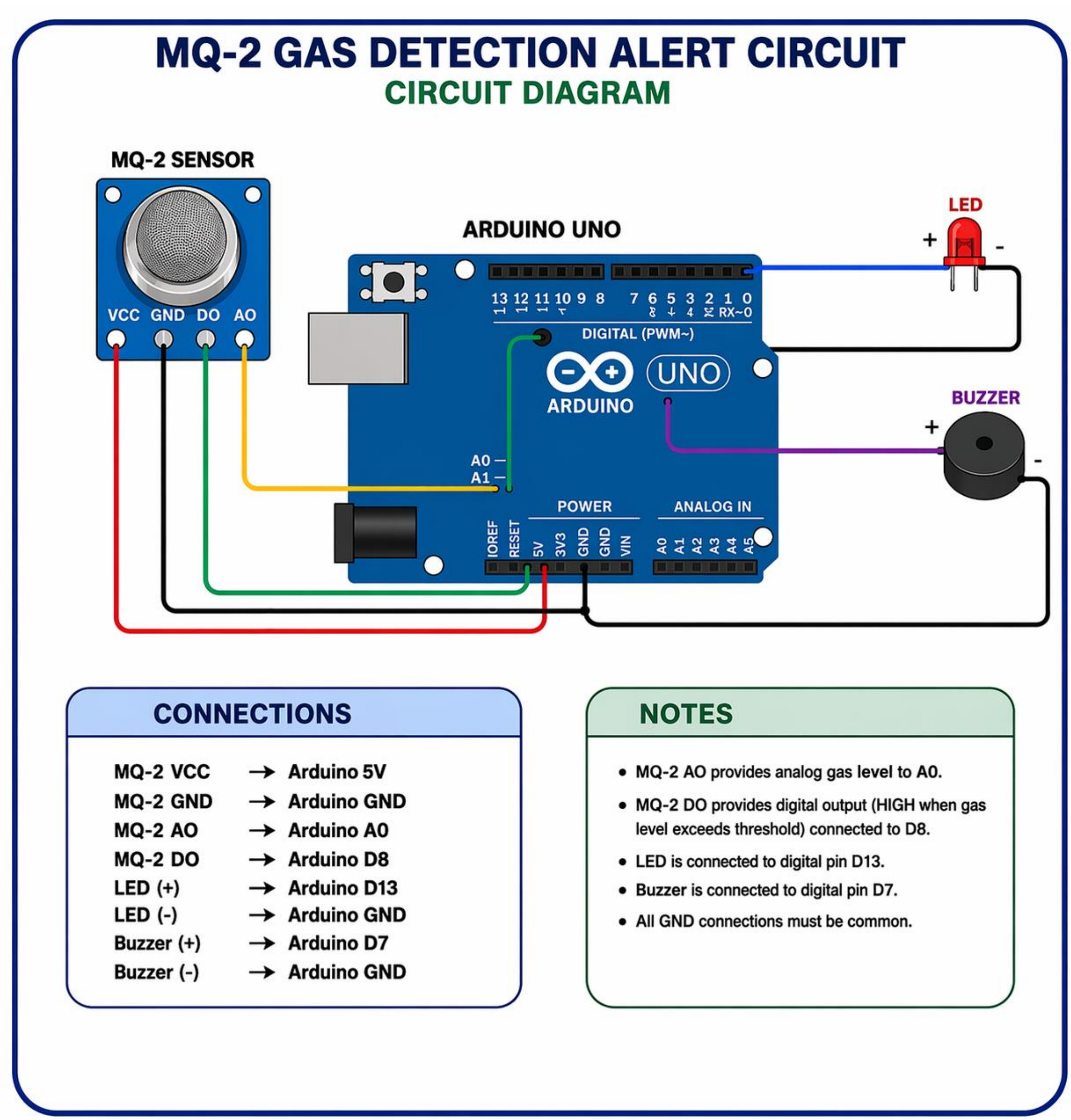
Advantages of Implementation

- Early gas leakage detection
- Improved safety
- Real-time monitoring
- Instant alert system
- Automatic device shutdown
- Low cost
- Easy installation
- Low maintenance
- Compact design
- Energy efficient

RESULT

The gas leakage detector system successfully identifies the presence of gas leakage and provides real-time monitoring of gas levels. When the gas concentration exceeds a predefined threshold, the system quickly activates an alarm or buzzer to alert users. It can also display gas levels and send notifications if integrated with

IoT modules. The system responds rapidly to leakage situations, helping to reduce the risk of fire and explosions. Overall, it offers reliable, accurate performance with low power consumption, making it suitable for both domestic and industrial safety applications.



CONCLUSION

The gas leakage detector system proves to be an effective and reliable solution for detecting hazardous gas leaks and ensuring safety. It provides quick and accurate detection, enabling timely alerts through alarms or notifications to prevent accidents such as fires and explosions. The system is cost-effective, energy-efficient, and easy to implement, making it suitable for both residential and industrial environments. Overall, the project demonstrates how technology can be used to enhance safety measures and minimize risks associated with gas leakage.

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